

Computational Neuroscience: Analysis of brain patterns using machine learning algorithms

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Abstract

This thesis investigates perceptual decision-making by combining computational neuroscience approaches with machine learning techniques applied to behavioral and electroencephalography (EEG) data. The study is based on an experiment in which participants repeatedly classify sequences of visual stimuli, allowing researchers to analyze how sensory evidence is integrated over time and how previous decisions affect the following ones.

In addition to data such as decisions and reaction times, EEG allows researchers to study brain activity related to the accumulation of evidence and previous decisions. The project aims to identify neural and behavioral patterns associated with the accumulation of sensory evidence and the confirmation bias, defined as the tendency to overvalue information consistent with previous decisions.

Supervised machine learning models will be trained to predict participants' decisions using behavioral and neural characteristics. The relative contribution of different variables will also be analyzed to determine which factors are most informative for prediction. The results will be compared with established computational models, such as the Diffusion Drift Model (DDM), to assess whether machine learning can capture similar or complementary patterns.

Overall, this work aims to contribute to the understanding of the cognitive and neural mechanisms behind perceptual decision-making and to evaluate the potential of machine learning as a tool for analyzing complex brain and behavioral data.

Keywords: Computational neuroscience, EEG, perceptual decision-making, machine learning, confirmation bias, sensory evidence accumulation, behavioral and neural analysis.

1 Problem definition and motivation

Decision-making is a fundamental cognitive process that involves the progressive integration of sensory information to reach a conclusion. In perceptual decision tasks, individuals accumulate sensory evidence over time until they reach a decision threshold. This process has been extensively studied using computational models such as the DDM, which provides a mathematical framework to describe how evidence is

integrated and how decisions are formed.

However, perceptual decisions are not entirely objective, as they can be influenced by previous decisions, leading to biases when processing new information. One of the most relevant phenomena in this context is confirmation bias, which refers to the tendency to give greater weight to information that matches with past decisions, while paying less attention to contradictory evidence.

Several studies suggest that this bias can emerge during the accumulation of sensory evidence, favoring information that supports previous decisions. Furthermore, confidence plays a significant role in the integration of new information, influencing whether people maintain or modify their decisions. Decisions made with greater confidence tend to be more difficult to change, while individuals with greater metacognitive ability, that is, a greater capacity to evaluate the accuracy of their own decisions, are more likely to detect errors and adjust their responses.

Although computational models such as the DDM have made significant progress, they rely on predefined assumptions about cognitive mechanisms. In contrast, machine learning techniques offer a more flexible, data-driven approach, capable of capturing complex patterns in behavior without imposing so many theoretical restrictions. However, their application to the study of perceptual decisions remains limited, especially when combined with neural data.

In this context, EEG allows the study of the brain activity related to decision-making processes. The collected data allows the measurement of brain activity with temporal precision, making it well-suited for studying how sensory evidence is processed and integrated over time. It is also important to note that EEG signals can reveal brain patterns related to the influence of prior decisions and confirmation bias.

Therefore, this project aims to combine behavioral and EEG data with machine learning techniques to investigate the decision-making process. Specifically, the goal is to identify patterns in neural data associated with the accumulation of sensory evidence and the influence of prior decisions. So, this study intends to evaluate whether machine learning models can capture the behavioral and neural characteristics of confirmation bias and provide complementary insights to traditional computational models.

Understanding these mechanisms has implications not only for neuroscience but also for fields such as artificial intelligence and clinical research.

1.1 Relevance of the project

Understanding how humans integrate sensory information when making decisions is a key aspect of cognitive neuroscience. While traditional computational models, such as DDM, have been useful in describing these processes, they rely on predefined assumptions about cognitive mechanisms.

In this context, combining behavioral and neural data, such as EEG, offers a more comprehensive approach to studying perceptual decisions. This allows direct observation of changes in brain activity over time, as well as the study of sensory

information processing and the influence of previous decisions on later neural responses.

At the same time, machine learning methods enable the analysis of complex data and can identify patterns without relying on prior assumptions. This makes them particularly useful for studying effects such as confirmation bias, where behavior depends on the interaction between sensory evidence and previous decisions.

The relevance of this project lies in integrating these approaches by applying machine learning techniques to experimental data in a perceptual decision task. This combination allows not only to predict participants' decisions but also to identify neural and behavioral patterns associated with the accumulation of evidence and the influence of previous choices.

In this way, the study contributes to the methodological development of computational neuroscience and provides new insights into the cognitive and neural mechanisms related to confirmation bias in human judgment.

2 Context and state of the art

Perceptual decision-making has been extensively studied in cognitive neuroscience and experimental psychology. In these tasks, people integrate sensory information over time before making a decision. This process is often explained by evidence accumulation models such as DDM.

However, decisions depend not only on current information but also on previous decisions. This effect is known as choice history bias, which reflects the tendency to repeat past responses or to interpret new information in a way that is consistent with them. One possible explanation for this is confirmation bias, a type of cognitive bias where people tend to give more weight to information that agrees with their previous decisions, while paying less attention to information that contradicts them. Furthermore, metacognition, especially the level of confidence, influences this process. Decisions made with greater confidence tend to be maintained, while greater metacognitive ability makes it easier to detect errors and adjust responses.

In recent years, experimental tasks with repeated stimuli have been used to study these effects, showing that prior decisions influence how new information is interpreted.

Although traditional models have been highly successful in describing perceptual decision-making, they rely on predefined assumptions about cognitive processes. Therefore, interest in machine learning techniques has grown, as they enable more flexible analyses of complex data. In this context, EEG is a key tool, providing access to brain activity with high temporal precision. The combination of behavioral data, EEG and machine learning offers a new way to study perceptual decision-making and better understand the cognitive and neural mechanisms involved.

3 Objectives

The main objective of this thesis is to analyze perceptual decision-making by combining behavioral and EEG data through the application of machine learning algorithms, in order to identify patterns associated with the accumulation of sensory evidence and detect potential cognitive biases in information integration.

Specifically, this work is based on the analysis of data obtained from an experiment in which participants were asked to categorize sequences of repeatedly presented visual stimuli. This experiment allows us to study the extent to which previous decisions influence the integration of new sensory information, as well as to analyze the role of other variables such as reaction time and neuronal activity measured by EEG.

From this general framework, the following specific objectives are defined:

Objective 1. To perform an exploratory analysis of the experimental data in order to identify relevant behavioral and neural variables involved in the decision-making process, including sensory evidence, participants' prior decisions, reaction times and EEG features.

Objective 2. To develop and train machine learning models capable of predicting participants' decisions.

Objective 3. To compare the performance of different machine learning algorithms in predicting participants' decisions, in order to determine which models achieve the best predictive performance.

Objective 4. To analyze the contribution of the different features used in the predictions, with the aim of identifying patterns consistent with the presence of confirmation bias in the integration of sensory evidence.

Objective 5. To study whether EEG activity differs when current information is consistent or not with previous decisions, to identify neural patterns of confirmation bias.

Objective 6. To compare the results obtained from the machine learning models with those reported in previous studies based on traditional computational models, such as the DDM, in order to assess whether machine learning approaches capture similar or complementary patterns.

Objective 7. To evaluate the potential of machine learning algorithms as a complementary tool for the study of cognitive and neural processes in the field of computational neuroscience.

4 Hypotheses

This project proposes the following hypotheses:

Hypothesis 1. Behavioral confirmation bias. Participants' decisions will be influenced by their previous responses, such that they will be more likely to repeat

prior decisions. In particular, sensory information consistent with previous decisions is expected to have a greater influence on current choices than contradictory information.

Hypothesis 2. Influence of confidence and decision stability. Higher confidence (or behavioral indicators such as faster reaction times) is expected to be associated with a greater likelihood of repeating previous decisions, reflecting more stable decisions and a lower impact of contradictory information.

Hypothesis 3. Neural correlates of decision-making and bias. EEG activity is expected to reflect the process of sensory evidence integration and may show differences depending on whether current information is consistent or inconsistent with previous decisions. This may suggest that information is processed in a biased way.

Hypothesis 4. Machine learning models predictions. Machine learning models are expected to accurately predict participants' decisions using behavioral and EEG data, indicating that these variables contain relevant information about their responses.

Hypothesis 5. Contribution of prior decisions to model predictions. Variables related to previous decisions are expected to have a significant contribution to model predictions. A strong influence of these variables would be consistent with the presence of confirmation bias in the integration of sensory evidence.

5 Methodology and work plan

To achieve the objectives of this thesis, a methodology based on the analysis of experimental behavioral and EEG data will be followed, combining techniques from computational neuroscience and machine learning in order to model and predict participants' judgments.

5.1 Experimental data

This work will be based on a dataset obtained from a perceptual experiment in which participants are required to evaluate sequences of visual stimuli and categorize them according to their mean orientation. In each experimental trial, participants make a categorical decision after observing a sequence of stimuli and the same sequence may be presented again in subsequent repetitions.

The dataset includes different types of data collected during the experiment. On the one hand, it contains participants' behavioral responses, such as their decisions and reaction times. On the other hand, it includes EEG recordings, which measure brain activity while participants perform the task.

This combination of data allows the joint analysis of behavioral responses and neural dynamics, making it possible to investigate how sensory evidence is processed and how prior decisions influence both behavior and brain activity.

5.2 Data preprocessing and exploratory analysis

In the first phase of the project, a data preprocessing procedure will be carried out in order to prepare the data for the use in the machine learning models. This stage will include tasks such as:

- data cleaning and validation
- organization of the experimental variables
- generation of new derived variables (for example, consistency with prior decisions or information about stimulus repetitions)

In addition, an exploratory data analysis will be performed in order to identify initial patterns in participants' behavior and to better understand the relationships between the different variables in the dataset.

5.3 Development of machine learning models

Once the data has been preprocessed, supervised machine learning models will be developed to predict participants' decisions.

Initially, interpretable baseline models such as logistic regression and support vector machines (SVM) will be implemented. These models allow the evaluation of linear relationships between the input features and the predicted decisions.

Subsequently, more flexible models based on non-linear approaches will be explored, including tree-based methods such as Random Forest and Gradient Boosting. These models are capable of capturing more complex interactions between behavioral and neural variables.

To ensure robust evaluation, the dataset will be divided into training and testing sets, and cross-validation techniques will be used to assess model generalization and reduce overfitting.

5.4 Model evaluation and interpretation

Once the models have been trained, a detailed analysis of the obtained results will be carried out. Model performance will be evaluated using standard classification metrics such as accuracy, precision, recall and F1 score. When appropriate, additional metrics such as the area under the ROC curve (AUC) may also be considered in order to provide a more complete evaluation of predictive performance.

The analysis will not only focus on accuracy, but also on how different variables influence the models, particularly previous decisions, stimulus characteristics and reaction times.

This will allow evaluating whether the predictive patterns identified by the models are consistent with the presence of confirmation bias in the process of sensory evidence integration.

6 Expected results

Based on the proposed hypotheses and methodology, several results are expected from the analysis of behavioral and brain activity data using machine learning techniques.

First, it is expected that participants' decisions will be influenced by their previous responses. In particular, decisions are likely to show a tendency to repeat previous choices, indicating the presence of confirmation bias in the integration of sensory evidence. This effect should be observable in the behavioral data as a higher probability of repeating previous choices.

Second, variables related to prior decisions and reaction times are expected to play an important role in predicting participants' responses. Faster reaction times, used as behavioral indicators of confidence, are expected to be associated with more stable decisions and a higher likelihood of repeating previous responses.

Third, EEG recordings are expected to reflect differences in brain activity depending on the relationship between current sensory evidence and previous decisions. In particular, differences may be observed between conditions in which the current evidence is consistent or inconsistent with prior responses, suggesting the presence of neural patterns associated with biased information processing.

From a machine learning perspective, it is expected that the developed models will be able to predict participants' decisions above chance level. This would indicate that both behavioral variables and EEG features contain relevant information about the decision-making process.

Furthermore, the analysis of feature importance is expected to show that variables related to prior decisions contribute significantly to model predictions. A strong influence of these variables would be consistent with the presence of confirmation bias.

Overall, the results are expected to show how previous decisions affect both behavior and brain activity, and to demonstrate that machine learning models can help identify patterns related to confirmation bias in perceptual decisions.

7 Project planning

The development of this thesis will be divided into different phases that will allow the project objectives to be addressed progressively. These phases include the literature review, the preprocessing and analysis of the experimental data, the development of machine learning models and the analysis of the obtained results.

Phase 1: Literature review and definition of the problem. During this first phase, a review of the scientific literature related to perceptual decisions, evidence accumulation models and cognitive biases, such as confirmation bias, will be conducted. Previous studies that use computational models and machine learning techniques in the analysis of human behavior will also be examined. This stage will make it possible to establish the theoretical framework of the project and to define more precisely the objectives and methodological approach of the study.

Phase 2: Data preprocessing and exploratory analysis. In this phase, the dataset obtained from the experiment will be analyzed. Data cleaning, structuring and preparation tasks will be carried out in order to make the data suitable for use in machine learning models. An exploratory analysis will also be performed with the aim of identifying initial patterns in participants' behavior and better understanding the relationships between the available variables.

Phase 3: Development of machine learning models. Once the data has been prepared, different machine learning models will be developed in order to predict participants' decisions based on the available experimental variables. Several classification algorithms will be implemented and trained, including models such as logistic regression, support vector machines and tree-based methods such as Random Forest or Gradient Boosting.

Phase 4: Model evaluation and analysis of results. In this phase, the performance of the developed models will be evaluated and compared in order to determine which algorithms achieve the best predictive performance. In addition, the importance of the different variables used in the predictions will be analyzed. This analysis will help identify possible patterns consistent with the presence of confirmation bias in the process of sensory evidence integration. The results obtained from the machine learning models will also be compared with those reported in previous studies based on traditional computational models.

Phase 5: Writing and revision of the final thesis. Finally, the thesis document will be written, including the description of the theoretical context, the methodology used, the results obtained and the discussion of the project's conclusions.

8 Project timeline – Gantt Chart

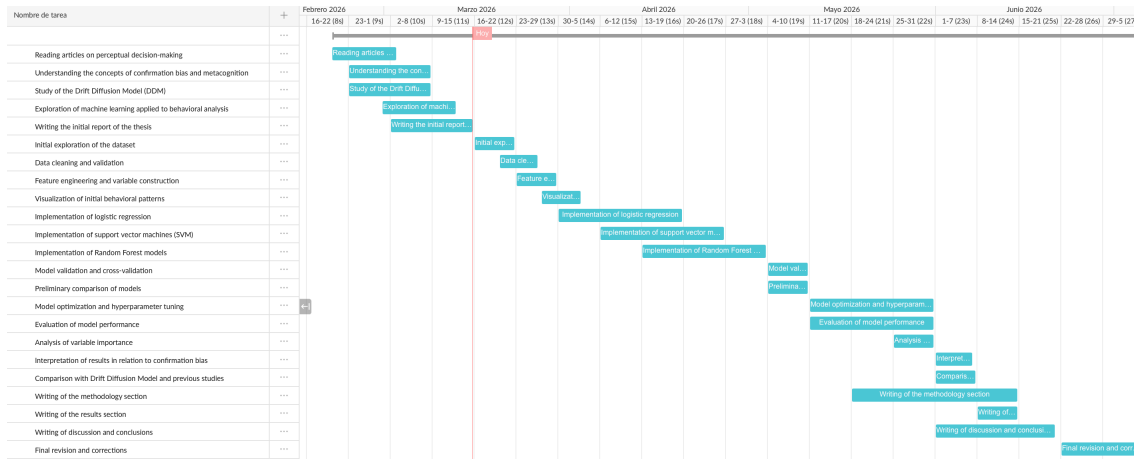


Figure 1: Gantt chart of the thesis development timeline.

Bibliography

References

- [1] Cichy, R. M., & Teng, S. (2017). Resolving the neural dynamics of visual and auditory scene processing in the human brain: A methodological approach. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 372(1714), 20160108. <https://doi.org/10.1098/rstb.2016.0108>
- [2] Heilbron, M., & de Lange, F. P. (2026). Higher-level spatial prediction in natural vision across mouse visual cortex. *PLOS Computational Biology*, 22(1), e1013136. <https://doi.org/10.1371/journal.pcbi.1013136>
- [3] Henin, S., Turk-Browne, N. B., Friedman, D., Liu, A., Dugan, P., Flinker, A., Doyle, W., Devinsky, O., & Melloni, L. (2021). Learning hierarchical sequence representations across human cortex and hippocampus. *Science Advances*, 7(8), eabc4530. <https://doi.org/10.1126/sciadv.abc4530>
- [4] Najafi, F., & Churchland, A. K. (2018). Perceptual decision-making: A field in the midst of a transformation. *Neuron*, 100(2), 453–462. <https://doi.org/10.1016/j.neuron.2018.10.017>

- [5] Parise, C. V. (2025). Correlation detection as a stimulus computable account for audiovisual perception, causal inference, and saliency maps in mammals. *eLife*, 14, RP106122. <https://doi.org/10.7554/eLife.106122>
- [6] Rollwage, M., Loosen, A., Hauser, T. U., Moran, R., Dolan, R. J., & Fleming, S. M. (2020). Confidence drives a neural confirmation bias. *Nature Communications*, 11, 2634. <https://doi.org/10.1038/s41467-020-16278-6>